

SCIENCE AND ELEMENTARY TECHNOLOGY SYLLABUS

LOWER PRIMARY (P1-P3)

KIGALI, 2026

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FOREWORD

The Rwanda Basic Education Board (REB) is honoured to present the second edition of the lower primary Science and Elementary Technology syllabus (P1–P3). This syllabus serves as an official guide to competence-based teaching and learning, providing coherence and consistency in the delivery of quality basic education across the country. Developed through the collective expertise of subject specialists, educators, and curriculum developers, it stands as a key resource in advancing the Competence-Based Curriculum (CBC) and reflects Rwanda’s continued commitment to nurturing knowledge, skills, values, and positive attitudes among learners.

Quality basic education is the foundation for Rwanda’s sustainable development and prosperity. In line with the country’s education philosophy, this syllabus emphasizes critical thinking, creativity, inclusiveness, and holistic learner development. It seeks to empower young people with the competencies necessary to achieve their full potential, equipping them with relevant knowledge, practical skills, and the right attitudes and values to integrate effectively into society and seize employment opportunities in a competitive world.

Anchored in Rwanda Vision 2050 and the National Strategy for Transformation (NST2), this syllabus supports the nation’s goal of building a knowledge-based economy driven by innovation, resilience, and global competitiveness. It prioritizes learner-centred pedagogy, active learning, and the practical application of knowledge to prepare pupils not only for academic success but also to become responsible, ethical, and productive citizens capable of contributing to Rwanda’s continued transformation.

This syllabus reflects our determination to equip learners with 21st-century skills, foster innovation, and instill values of integrity, social responsibility, and lifelong learning. Its success depends largely on the dedication and professionalism of teachers and school leaders. We therefore call upon all educators to embrace the approaches, methodologies, and assessment strategies outlined herein with commitment and passion, transforming classrooms into dynamic spaces of curiosity, creativity, and inspiration.

On behalf of REB, I extend my heartfelt appreciation to all who contributed to the development of this syllabus. Your expertise and dedication have been instrumental in shaping a curriculum that responds to the needs of our learners and supports Rwanda's national aspirations. Together, we can ensure that education continues to be a powerful driver of our nation's development and prosperity.



Dr. MBARUSHIMANA Nelson

Director General, REB

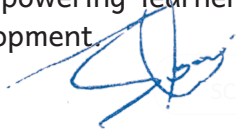
ACKNOWLEDGEMENT

The development of this syllabus has been a collective effort that reflects the dedication, expertise, and commitment of many stakeholders. On behalf of the Curriculum, Teaching and Learning Resources Department of the Rwanda Basic Education Board (REB), I wish to express my deepest appreciation to all who contributed to the successful preparation of this lower primary Science and Elementary Technology syllabus (P1–P3).

Special thanks go to the subject experts, curriculum developers, teachers, and education partners whose insights, feedback, and professional contributions have been invaluable in shaping this syllabus. Your collaboration has ensured that the syllabus is not only aligned with Rwanda’s Competence-Based Curriculum but also responsive to the aspirations of Vision 2050 and the National Strategy for Transformation (NST2).

We also acknowledge the invaluable support of development partners and education stakeholders who have consistently provided technical assistance in curriculum adaptation. Your partnership has greatly contributed to strengthening the quality of Rwanda’s education system.

Finally, we call upon teachers, school leaders, and all education practitioners to use this syllabus with commitment and professionalism, ensuring that classrooms become vibrant learning environments that nurture knowledge, skills, values, and positive attitudes among learners. Through your dedication, this syllabus will fulfil its purpose of empowering learners to realize their full potential and contribute meaningfully to Rwanda’s sustainable development.



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1.INTRODUCTION

1.1. Background to syllabus adaptation

The adaptation of the Science and Elementary Technology (SET) syllabus for lower primary reflects Rwanda's ongoing commitment to strengthening the Competence-Based Curriculum (CBC), first rolled out in 2015. Continuous feedback from teachers, school leaders, and education partners highlighted the need for improvements in content delivery, time allocation, and responsiveness to learner needs.

This second edition incorporates the Ministerial directive to harmonize school hours to five per day and 33 weeks of learning annually and ensures greater coherence, inclusiveness, and alignment with national priorities such as *Vision 2050* and the *National Strategy for Transformation (NST2)*, which emphasize a knowledge-based economy, responsible citizenship, and sustainable development.

2. TEACHING AND LEARNING SCIENCE AND ELEMENTARY TECHNOLOGY

2.1.The rationale of teaching and learning SET

2.1.1.Science and Elementary Technology and society

Teaching Science and Elementary Technology to young children is critical for establishing a foundation for further success in science and for coping with the demands of the 21st century. Furthermore, technology education constitutes an unequalled important added value. Not only in developed countries but also in developing countries such as Rwanda, the love and interest in science and technology begins in primary school where young children tend to be more curious and motivated to learn.

The inclusion of Science and Elementary Technology in the Primary School reflects the importance of science and technology in many aspects of our daily lives, at work, at school and at home. As integrated science, it provides a

very good foundation for the study of science subjects in the post-primary setting. Most importantly, it cultivates a positive attitude towards science and provides pupils with opportunities to experience the excitement of working as a scientist.

Above all, the rationale of teaching and learning of SET is embedded in the need for pupils to have a greater awareness of the role of science and technology in everyday life. SET at primary school enables the pupil to develop competencies which have great impact on the society in general.

2.1.2.Science and Elementary Technology and pupil

Pupils have to be prepared from an early age for active and responsible citizenship. With this regard, SET strives to equip pupils to understand and situate scientific and technological developments in their cultural, environmental, economic, political and social contexts. At the center of teaching and learning of SET, hands on activities will play a key role, which in turn, should contribute significantly towards improving pupils' achievement, motivation, technological literacy and test scores.

2.2.Competences

A competence is the ability to use an appropriate combination of knowledge, skills, attitudes, values, and behaviours to accomplish a task successfully. In this syllabus, competences reflect a holistic blend of these elements, enabling learners to apply them confidently and appropriately in diverse real-life situations.

By acquiring these competences, learners become better equipped to handle challenges, make informed decisions, act responsibly, and contribute positively as active members of society. The competences are embedded in the broad subject competences, yearly objectives, and unit-specific learning competencies. Both generic competences and basic competences must be emphasized and consistently reflected in the teaching and learning process.

Teachers are encouraged to design learning experiences that engage learners in meaningful tasks, allowing them to practice and demonstrate competences both during and at the end of the learning process.

2.2.1. Generic competences

- **Critical and problem solving skills:** Pupils develop the ability to think broadly and objectively, evaluate their causes and consequences, and suggest peaceful and practical solutions that contribute to harmony and sustainable living in their communities.
- **Creativity and innovation:** The acquisition of such skills will help pupils to take initiative and use imagination beyond the knowledge provided, to generate new ideas and construct new concepts.
- **Research:** This will help pupils to find answers to questions based on existing information and concepts and use it explain phenomena from gathered information.
- **Communication in official languages:** Teachers, irrespective of not being teachers of language, will ensure the proper use of the language of instruction. This will help learners develop the capacity to communicate clearly, confidently, and respectfully in both spoken and written forms. By using appropriate vocabulary and structures in social, economic, political, religious, and cultural contexts, they engage in meaningful dialogue, debates, and discussions that promote coherent and logical exchange of ideas.
- **Cooperation, interpersonal management and life skills:** This competence helps learners to work collaboratively with others, respect diverse opinions, and demonstrate empathy in group tasks. It also cultivates positive moral values, responsibility, and participation in practical activities such as environmental protection, health promotion, community service and responding creatively to the variety of challenges encountered in life.
- **Lifelong learning:** Learners cultivate curiosity and adaptability by linking classroom knowledge to real-life experiences. This equips them with the ability to update knowledge and skills independently, cope with evolving societal needs, and pursue personal growth and continuous development.

2.2.2. Science and Elementary Technology broad competences

The teaching of Science and Elementary Technology to pupils for primary school aims to:

- develop an interest in science as a body of knowledge and methods of thinking, inquiring and working,
- develop a concern for the world around us,
- make children aware of themselves within the world we live and the importance of science in relation to this,
- help children realize the importance of technology for society and the need for scientific knowledge to understand the present technological age,
- Develop positive attitudes, practical skills and basic scientific concepts.

The overall objective of learning SET at primary level is to promote basic scientific literacy. Upon completion of primary education, pupils will have acquired competences (knowledge, skills and attitudes) which will enable them to:

- demonstrate an understanding of the nature of science,
- apply inquiry-based science learning strategies,
- solve problems in their daily life,
- reflect good habits of personal hygiene and their surroundings,
- practice good habits to protect their own health and that of others,
- reflect a positive attitude towards environmental protection and conservation,
- improve crop and animal husbandry,
- demonstrate an understanding of the interrelationship between science, society and technology,
- Perform SET learning through ICT and make based inquiry projects in the daily life.

2.2.3.Science and Elementary Technology and developing competences

National policies highlight the need to develop competences that foster higher-order thinking and guide learners

to become responsible, virtuous, and engaged citizens. These competences are built through interactive activities such as discussions, group work, presentations, role-plays, case studies, and practical problem-solving.

Through Science and Elementary Technology, learners acquire critical thinking, analytical skills, observation, research, communication, interpersonal awareness, intercultural understanding, and cooperation. These skills enable them to navigate challenges, participate responsibly in society, and express ideas, argue, and debate logically and respectfully.

Science and Elementary Technology encourages pupils to manipulate different materials and data during class experiments and to undertake project work that involve analytical and problem-solving skills directed towards innovation, creativity and research activities by pupils.

The acquired knowledge in learning Science and Elementary Technology should develop a responsible citizen who adapts to scientific reasoning and attitudes and develops confidence in reasoning independently. The pupil should show concern of individual attitudes, environmental protection and comply with the scientific method of reasoning. The scientific method should be applied with the necessary rigor, intellectual honesty to promote critical thinking while systematically pursuing the line of thought.

2.3. Values

Values are enduring beliefs, principles, and dispositions that influence how individuals think, feel, make decisions, interact with others, and apply knowledge, skills, and attitudes in different situations. Within the Competence-Based Curriculum, values provide the ethical, social, and cultural foundation for learning and responsible citizenship. The Rwanda Curriculum Framework recognizes two complementary categories of values: Curriculum Values, which guide the philosophy, design, and implementation of education, and Basic Values, which represent the moral, civic, and character attributes that learners are expected to internalize and

demonstrate in their daily lives. Together, these values shape the development of competent, responsible, self-reliant, and socially engaged citizens.

2.3.1 Curriculum Values

- **Excellence:** Striving for the highest standards in learning, behaviour, and performance by consistently producing quality work, using feedback for improvement, and taking pride in achievement. It is reflected in accuracy, continuous growth, and encouraging excellence in others.
- **Aspiration and Optimism:** Believing in potential and pursuing ambitious goals with confidence, perseverance, and a positive outlook. It involves viewing challenges as opportunities for growth and inspiring hope in oneself and others.
- **Equity and Inclusiveness:** Valuing the dignity and worth of every individual by promoting fairness, equal participation, and opportunities for all. It is demonstrated through respect for diversity, support for others, and opposition to discrimination and exclusion.
- **Openness and Transparency:** Acting honestly, ethically, and accountably while fostering trust and participation. It is reflected in clear communication, responsible decision-making, openness to feedback, and fairness in interactions with others.
- **Importance of Family, Rwandan culture and heritage:** Recognizing the role of family, culture, and heritage in shaping identity, values, learning, and social development. It is demonstrated through respect for family relationships, participation in community life, and appreciation of cultural traditions and

responsibilities.

- **Learner-centredness:** Placing learners at the heart of education by recognizing their needs, interests, abilities, and experiences while promoting active engagement and ownership of learning. It is reflected in inclusive, supportive learning environments where learners participate, explore, collaborate, reflect, and take responsibility for their learning.

2.3.2 . Basic Values

- **Dignity and Integrity:** Upholding self-worth and acting consistently according to moral and ethical principles. It is demonstrated through honesty, responsibility, respect for self and others, and doing what is right even when unsupervised.
- **Self-Reliance:** Taking responsibility for one's learning, decisions, and well-being through initiative, confidence, and independent action. It involves solving problems, managing resources effectively, and making informed choices.
- **National and Cultural Identity:** Appreciating and promoting Rwanda heritage, values, language, traditions, and national aspirations. It is reflected in national pride, respect for cultural heritage and institutions, and commitment to unity and social cohesion.
- **Peace and Tolerance:** Living harmoniously with others by respecting differences and resolving conflicts constructively. It is demonstrated through empathy, respectful dialogue, reconciliation, and support for peaceful coexistence.
- **Justice:** Promoting fairness, equity, and respect for rules, rights, and responsibilities. It is reflected in impartial decision-making, ethical judgement, consistent application of rules, and opposition to unfair treatment.
- **Respect for Others and Human Rights:** Recognizing and upholding the dignity, rights, and freedoms of every

person. It is demonstrated through courtesy, active listening, inclusion, respect for diversity, and protection of the rights of others.

- **Solidarity and Democracy:** Working collaboratively for the common good while respecting participation, shared responsibility, and collective decision-making. It is reflected in teamwork, community service, social responsibility, and respect for diverse views.
- **Patriotism:** Demonstrating love, loyalty, and commitment to the wellbeing and development of Rwanda. It is expressed through civic responsibility, community participation, promotion of national unity, and positive representation of the country.
- **Hard Work:** Applying consistent effort, diligence, and responsibility in pursuing goals and fulfilling duties. It is demonstrated through persistence, productive use of time, initiative, and commitment to quality outcomes.
- **Commitment and Resilience:** Remaining dedicated to goals and continuing to make progress despite difficulties and setbacks. It is reflected in perseverance, adaptability, learning from mistakes, emotional self-control, and determination to succeed.

3. Pedagogical approach

The pedagogical approach in Science and Elementary Technology under the CBC emphasizes learner-centered, participatory, and inclusive methods that develop knowledge, skills, values, and competences for responsible citizenship. Teaching is interactive and active, engaging learners through debates, discussions, role-plays, storytelling, field visits, projects, research, case studies, and reflective activities. These approaches ensure that learning is enjoyable, personalized, and connected to real-life situations, while also fostering critical thinking, moral reasoning, and problem-solving.

3.1. Role of the teacher

Teachers act as facilitators and guides, shifting away from traditional rote instruction. They organize learners

individually, in pairs, or in groups, and design purposeful tasks that promote inquiry, collaboration, and autonomy. Teachers apply pedagogical principles of progression—moving from simple to complex, concrete to abstract, and from learners’ experiences to broader knowledge rooted in social, moral, and religious contexts. They integrate cross-cutting issues such as peace, gender equality, environmental conservation, health, and cultural values, while ensuring sensitivity to learners’ diverse backgrounds. In Religious Education, teachers adopt an inclusive approach by presenting different faiths objectively, encouraging respect, understanding of diverse beliefs, and appreciation of moral and spiritual values.

3.2. Role of the pupil

Learners are active participants and co-creators of knowledge. Simply, it emphasizes the need for children to think about scientific activity in order to make sense of and understand the scientific concepts being introduced. Specifically, when engaging in inquiry, pupils will describe objects and events, ask questions, construct explanations, test those explanations against current scientific knowledge, and communicate their ideas to others. In SET lesson, pupils get involved in the following activities (IAP, 2010, p. 9):

- observing and, where possible, handling and manipulating real objects;
- pursuing questions which they have identified as their own even if introduced by the teacher;
- taking part in planning investigations with appropriate controls to answer specific questions;
- using and developing skills of gathering data directly by observation or measurement and by using secondary sources;
- using and developing skills of organizing and interpreting data, reasoning, proposing explanations, making predictions based on what they think or find out;
- working collaboratively with others, communicating their own ideas and considering others’ ideas;
- expressing themselves using appropriate scientific terms and representations in writing and talk;

- engaging in lively public discussions in defense of their work and explanations;
- applying their learning in real-life contexts;
- reflecting self-critically about the processes and outcomes of their inquiries.

During this reciprocal interaction, what pupils will acquire is not only content knowledge, but a number of skills including how to approach a problem, identify important resources, design and carry out hands-on investigations, analyze and interpret data, and, perhaps most importantly, recognize when they have answered the question or solved the problem (Marsha, 2000).

3.3. Inclusive and special needs education

In line with Rwanda's inclusive education policy, all learners regardless of ability have the right to access quality education. Teachers adapt teaching methods, materials, and assessment strategies to meet diverse needs, including those of learners with disabilities. Differentiated instruction, assistive technologies, peer support, and individualized approaches are used to ensure equitable participation and achievement of learning outcomes. This inclusive approach recognizes learners' varied paces, experiences, and abilities, creating an environment where every learner can thrive.

4. Assessment approach

Assessment within the CBC is a fundamental component of the teaching and learning process. It is designed to evaluate learners' progress across knowledge, skills, attitudes, and values, while simultaneously supporting continuous learning and development. Unlike traditional models, CBC assessment emphasizes real-life problem-solving, requiring learners to apply acquired competences to complex, everyday situations.

Assessment entails the systematic collection and interpretation of evidence to measure learner performance against established standards. Beyond measuring achievement, it informs instructional improvement and

provides constructive feedback to learners, parents, and educators, thereby fostering accountability and guiding educational decision-making.

4.1. Levels of assessment across terms and national examinations

To provide a comprehensive picture of learner achievement and development, assessment is conducted at multiple levels:

- i. **Diagnostic:** Conducted before instruction to identify learners' prior knowledge and skills; administered at the beginning of a unit or lesson to guide teaching.
- ii. **Formative / Continuous:** Carried out during the learning process to track progress and provide feedback. They are implemented continuously through quizzes, projects, class activities, and teacher observations.
- iii. **Summative assessment:**
 - **End Term I:** Administered after instruction to evaluate learners' achievement of Term I content. They are conducted at the end of the first term as a school-based assessment.
 - **End Term II:** District-coordinated assessment designed to measure achievement of Term II content. They developed by schools and quality-checked at district level.
 - **End Term III:** National assessment to evaluate mastery of the entire year's content. They are prepared, set, and moderated by NESA at the end of the third term.

National Examinations (NESA): High-stakes assessments for certification, selection, and accountability. They are administered nationally at the end of key education cycles (P6, S3, S6) under NESA's authority. This examination integrates objective questions, structured responses, and applied tasks aligned with CBC principles to holistically measure learner achievement.

Average scores from school, district, and comprehensive summative assessments will be weighted and included

in the final national examination grading, initially contributing 10%, with the percentage gradually increasing as teachers gain experience and confidence in assessment practices.

4.2. Record Keeping

Accurate record keeping is essential for monitoring learner progress and informing instructional decisions. Teachers use tools such as portfolios, checklists, and report books to collect evidence of learning. Portfolios include samples of learners' work, test results, projects, and self-reflections, serving as both a record of achievement and a diagnostic tool for guiding remedial action. These records provide data for feedback to learners, teachers, and parents, and contribute to final evaluations.

4.3. Item writing in summative assessment

Summative assessment items must be valid, fair, reliable, and aligned with CBC principles. They should test a range of competencies, including knowledge recall, application, analysis, and values integration, while reflecting real-life contexts. Item writers follow a structured process that involves:

- Identify topic areas to be tested from the subject syllabus.
- Outline subject matter content to be considered as the basis for the test.
- Identify learning outcomes to be measured by the test.
- Preparing a test blueprint or table of specifications;
- Using verbs that require higher-order thinking beyond memorization;
- Applying transparent and consistent marking schemes.
- Greater weight is given to higher-order cognitive skills as stated in the syllabus.

4.4. Structure and format of national examinations

At the completion of primary education, learners undertake a national examination designed to evaluate literacy, numeracy, and competences in core subjects. For Science and Elementary Technology, a single combined paper is administered, integrating both subject areas. The examination employs a variety of item types—such as objective questions, structured responses, and applied tasks—intended to holistically assess both subject knowledge and broader CBC competencies.

4.5. Reporting to parents

Regular communication with parents is an essential component of the CBC. This is achieved through progress reports, parent-teacher meetings, and school community activities. Since the CBC emphasizes a wide range of learning outcomes: knowledge, skills, attitudes, and competences—a single mark is insufficient to capture learner progress. Effective reporting should highlight areas where learners are excelling as well as those requiring improvement. Using descriptive feedback or simple rating scales (e.g., meeting expectations very well, meeting expectations, not meeting expectations) across knowledge, subject-specific skills, and competences provides a clearer and more comprehensive picture than numerical grades alone. This approach fosters meaningful parental involvement and supports learners' continued growth.

5. Resources

The effective implementation of the Science and Elementary Technology syllabus requires the provision and utilization of appropriate resources. These resources include both teaching and learning materials and qualified human personnel, without which the delivery of the syllabus would be ineffective.

5.1. Materials needed for implementation of this syllabus

A variety of materials are essential to facilitate teaching and learning processes in Science and Elementary Technology. These materials provide learners with practical experiences, stimulate critical thinking, and foster the acquisition of competences. They include:

- **Textbooks and reference materials:** Up-to-date textbooks, teacher's guides, learner's books, dictionaries, and foundational religious texts such as the Bible, the Qur'an, and other relevant doctrinal sources.
- **Teaching and learning aids:** Visual aids such as science kit items, improvised teaching aid, ICT equipment including the internet network
- charts, posters, flashcards, diagrams, maps, and models; audio-visual aids including projectors, radios, and videos; and digital resources such as e-books, educational websites, and online learning platforms that enhance conceptual understanding.
- **Classroom materials:** Chalkboards/whiteboards, flip charts, markers, and stationery to support daily classroom instruction.
- **Library and resource centre materials:** Access to a well-equipped library with supplementary reading materials, journals, and reference documents on history, culture, ethics, and religion.
- **Field visit resources:** Materials to support educational tours and fieldwork, such as journals, notebooks, and logistical support, since field visits and community interactions are integral to experiential learning in this subject.
- **Community and cultural resources:** Access to local leaders, religious institutions, museums, cultural heritage sites and genocide memorial sites that provide learners with real-life exposure to the values and practices being studied.
- **Assessment tools:** Rubrics, checklists, portfolios, and test materials are necessary to support continuous and summative assessment in line with the competence-based approach.

These resources should be adapted to the learners' context, ensuring inclusivity and accessibility for learners with special educational needs.

5.2.Human resource

The successful implementation of this syllabus relies heavily on qualified, competent, and motivated human resources. The Science and Elementary Technology teacher should be equipped with a strong pedagogical content knowledge (PCK) and should be creative and able to improvise since many of teaching aids can be found around the school and hand-made by the teachers themselves.

5.3.Skills required for the teacher of SET

Effective teaching of Science and Elementary Technology requires teachers to possess a broad set of professional, pedagogical, and interpersonal skills. These competencies enable them to deliver accurate subject content, promote values and attitudes, and foster holistic learner development in line with the Competence-Based Curriculum. The essential skills include:

a) Subject matter expertise

- Strong mastery of science concepts and technological practices to ensure accurate and comprehensive content delivery.
- Ability to make meaningful connections between Science and Elementary Technology and other disciplines, linking classroom learning to real-life contexts.

b) Pedagogical and instructional skills

- Capacity to design and implement learner-centered, participatory, and inclusive lessons using debates, role-plays, storytelling, projects, and discussions.

- Competence in integrating cross-cutting issues such as peace, gender, human rights, and environmental sustainability.
- Creativity and innovation in developing and applying diverse teaching resources, including ICT tools, realia, and community-based materials.
- Ability to adjust instruction to learners' levels, ensuring inclusivity and engagement for all.

c) Assessment skills

- Proficiency in designing and applying formative and summative assessments aligned with curriculum competencies.
- Effective use of rubrics, portfolios, checklists, and performance tasks to monitor knowledge, skills, and values acquisition.
- Skill in giving constructive feedback that motivates learners and guides improvement.

d) Classroom management and facilitation

- Ability to establish a respectful, safe, and inclusive learning environment that promotes tolerance and values diversity.
- Proficiency in managing group work, discussions, and collaborative Learning activities.
- Sensitivity and neutrality in addressing religious, ethical, and cultural issues.

e) Communication and interpersonal skills

- Strong oral and written communication to simplify complex ideas, stimulate dialogue, and foster critical reflection.
- Active listening and empathy to respond effectively to learners' academic and personal needs.
- Collaboration with colleagues, parents, school leaders, and community stakeholders to support holistic learner growth.

f) Guidance and counseling skills

- Ability to provide moral, spiritual, and emotional guidance to learners, supporting their ethical development and social well-being.
- Demonstrating passion, impartiality, and genuine care for children in the teaching and learning process.

g) Professional and ethical competence

- Commitment to continuous professional development through reflective practice, training, and peer collaboration.
- Integrity, impartiality, and ethical conduct, serving as a role model for learners.
- Capacity to balance academic instruction with the cultivation of values and responsible citizenship.

h) Research and innovation skills

- Engagement in reflective practice and classroom-based research to improve teaching and learning.
- Adaptation of innovative approaches and ICT tools to enhance learner participation.
- Ability to contextualize global knowledge within Rwanda's socio-cultural environment.

6. SYLLABUS UNITS

6.1. Presentation of the structure of the syllabus units

The Science and Elementary Technology syllabus is designed as a single, integrated syllabus implemented uniformly across all grades. Its structure follows a logical progression that ensures clarity, consistency, and alignment with the Competence-Based Curriculum approach.

At each grade level, the syllabus is organized into **Topic areas** and each topic area is composed of **Units**, and

every unit is characterized by the following features:

1. **Alignment with lessons:** Each unit corresponds to a specified number of lessons, ensuring effective time allocation and instructional planning.
2. **Key unit competence:** Every unit is built around a *Key unit competence* that guides all teaching and Learning activities. This competence represents the overall learning goal to be achieved by both teacher and learners.
3. **Learning objectives:** Each key unit competence is translated into specific Learning objectives, categorized as follows:
 - **Type I (Knowledge and Understanding):** Focused on foundational knowledge and comprehension, often referred to as *Lower Order Thinking Skills (LOTS)*.
 - **Type II and Type III (Skills, Attitudes, and Values):** Aimed at developing practical skills, critical thinking, ethical values, and positive attitudes. These objectives, known as *Higher Order Thinking Skills (HOTS)*, form the core focus of the adapted competence-based curriculum.
4. **Content specification:** Each unit outlines the scope of content to be taught and learned, ensuring that teachers and learners work within the boundaries set by the stated Learning objectives.

6.2. Science and Elementary Technology for Primary One (P1)

6.2.1 General competences

At the end of P1, pupil should achieve the following main competences:

- Identify, name, use and keep properly materials used at home and at school.
- Identify various ICT tools and their uses, and use radio, and television correctly.
- Name where water comes from and why water is important.
- Identify domestic, wild and aquatic animals and their importance.
- Name plants that people grow and plants that grow on their own.
- Name the main parts of the human body, describe what they do, and keep them clean.

6.2.2. Table units for Primary 1

Topic area: Elementary technology					
Science and Elementary Technology: Primary one			Unit 1: Materials used at school and at home		Number of periods: 10
Key unit competence: Identify, name, use and keep properly materials used at home and at school.					
Learning objectives			Content	Learning activities	
Knowledge and understanding	Skills	Attitudes and values			
– Name some items that are used at home and at school. – State what school items are used for. – State what home items are used for. – List the items used for cleaning at home.	– Group materials used at school and at home basing on their uses. – Use properly materials used at home and at school. – Clean and keep properly things used at school. – Clean and keep properly things used at home.	– Take care of things used at home and school. – Put things back in the right place after using them. – Show responsibility while using home and school materials.	– School materials and their uses: • Writing materials (pens,pencils, notebooks, chalks, a chalkboard, chalks and papers). • Drawing materials (pencils, drawing pads/ books, rubber, sharpener). • Reading materials (Newspapers, books). • Time telling objects (Clock, sun, shadow, and watch).	– Observe and name materials used at school. – Touch various materials used at school. – Group discussions or brainstorm to find out the names and uses of materials used at school. – Clean classroom and various school materials.	

		- Be careful while using home and school materials. - Willingness to share materials with others.	• Storage materials (school bag, locker, desk, drawer). • Cleaning materials at school (soap, water, duster, broom, mopping rag and basin...). - Home material and their uses: • Kitchen materials and dining materials (cooking pots, knives, cups, spoons, forks, and plates ...). • Sitting room materials (chairs, table, television). • Bedroom materials (bed, mattress, covers, mosquito net, table, shelves...). • Home storage materials (store, cupboard, boxes, shelves, tables...). • Home cleaning materials (soap, water, basin, broom, sponges, and others.).	- Store various school materials. - Observe and touch various tools and materials used in their homes. - Group discussions or brainstorm to discover the names and use of tools and materials used at home. - Clean various tools and materials used at home. - Discuss how to store the home materials. - Draw and shade some school and home materials. - Make a spoon using local materials through engineering process.
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Links to other subjects: *Mathematics (grouping items), English (learning vocabulary), Social and Religious Studies (cleanliness and responsibility), Creative Arts (drawing materials).*

Generic competences:

- **Communication (Literacy and Numeracy):** Developed through naming, discussing, and presenting the uses of school and home materials during group discussions and classroom activities.
- **Co-operation, interpersonal relations and life skills:** Developed through working in pairs or groups while observing, cleaning, organizing, storing, and sharing school and home materials.
- **Critical thinking:** Developed through comparing, classifying, and grouping materials according to their uses and explaining the reasons for their classification.
- **Creativity and innovation:** Developed through drawing, colouring, and suggesting appropriate ways of using, caring for, and storing school and home materials.

Basic values:

- **Dignity and integrity:** Developed through using materials responsibly, taking care of them, and returning them to their proper places after use.
- **Respect for others and human rights:** Developed through sharing learning materials, respecting others' property, and using common materials carefully.
- **Hard work:** Developed through active participation in cleaning, organizing, and maintaining school and home materials.
- **Solidarity and democracy:** Developed through teamwork, cooperation, and helping one another during classroom and practical activities.

Cross-cutting issues

a) Gender education

- Involve both girls and boys in the activities of the lesson.
- Show that all pupils, boys and girls, can use any material.

b) Environment and sustainability

- Teach pupils to keep materials clean and avoid throwing rubbish on the ground.

c) Peace and values education

- Promote respect and sharing of materials with others.

Assessment criteria: *Assess pupils' ability to identify, name, and use properly school and home materials, correctly group them by function or place, and demonstrate care and respect for the materials.*

Materials: *Pen, pencil, coloured pencils, chalk, coloured chalk, painting brush and paint, paper, a blackboard, a cupboard, a locker, a duster, water, manila paper, books, chairs, learning aids and materials, a table, plates, dishes, forks, cups, pans, a pounder/mortar, spoons, brooms, a basket, a mop, toilet paper, clothes, a knife, an iron bar, soap, boxes,...*

Topic area: ICT

Science and Elementary Technology:
Primary one

Unit 2: ICT tools

**Number of
periods: 10**

Key unit competence: Identify various ICT tools and their uses and use them correctly.

Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Name different ICT tools. – Identify different ICT tools. – State what ICT tools are used for. – Demonstrate how to use radio and television.	– Match ICT tools with their correct uses. – Turn on and turn off radio. – Show how to use a radio by changing channels. – Turn on and turn off television. – Show how to use a television and change channels.	– Appreciate the importance of ICT tools in everyday life. – Show interest and curiosity in using radio and television. – Take care of ICT tools. – Be careful when using a phone or television. – Use ICT tools when needed.	– Names of ICT tools: Radio, Telephone, Computers, Television, Loudspeakers, Microphone – Uses of ICT tools: Listening, calling, typing, playing, watching, making sounds louder. – How to use ICT tools Using a radio • Using a television	– Observe and identify different ICT tools. – Brainstorm to name different ICT tools. – Draw and shade different ICT tools. – Discuss and brainstorm about the uses of different ICT tools.

				– Practice on how to turn on and turn off a radio. – Practice on how to change radio channels and adjust the volume. – Practice on how to turn on and turn off a television. – Practice on how to change television channels and adjust the volume.
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Links to other subjects: *Mathematics (counting ICT tools), English (vocabularies), Social and Religious Studies (importance of communication), Creative Arts (drawing and coloring ICT tools)*

Generic competences:

- **Communication (Literacy and Numeracy):** Developed through naming ICT tools, discussing their uses, and explaining how to use radio and television during class activities.
- **Critical thinking:** Developed through matching ICT tools with their correct uses and selecting the appropriate tool for different everyday situations.
- **Lifelong learning:** Developed by encouraging pupils to appreciate the usefulness of ICT tools and develop confidence in using them responsibly in their daily lives.

Basic values:

- **Dignity and integrity:** Developed through handling ICT tools responsibly, following instructions, and taking care of the equipment during practical activities.
- **Self-reliance:** Developed through practicing how to operate a radio and television independently and using ICT tools appropriately when needed.

Cross-cutting issues

- **Environmental and sustainability:** Tell the pupils that they must avoid disposing off badly the e-waste and emphasize the importance of proper disposal of electronic devices like batteries or broken radios.
- **Financial education:** Encourage pupils to switch off ICT tools when not in use so as to use well the electricity

Assessment criteria: Assess pupils' ability to name ICT tools, state their uses, demonstrate how to use radio and television properly, and show care and respect for the tools.

Materials: Radio, television, telephone, computer/laptop, loudspeaker, microphone, camera, pictures or videos of various ICT tools, radio batteries,

Topic area: Environment

Science and Elementary Technology:
Primary one

Unit 3: Water

Number of periods: 10

Key unit competence: Name where water comes from and why water is important.

Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Name different places where water comes from. – Outline the types of water. – Point out why water is important. – Recall the steps to wash things	– Match each source of water with a picture or example. – Classify water as clean or dirty by looking at its appearance. – Wash the handkerchief properly. – Wash hand towels and socks. – Clean vegetables and other foods before cooking or eating.	– Show interest in knowing where water comes from. – Take care of water sources. – Keep clothes clean and neat. – Always wash raw food before eating and remind others to do the same. – Share water fairly with others.	– Sources of water – Taps, rain, rivers, lakes – Types of water – Clean water, dirty water – Importance of water: drinking, cleaning, cooking, washing, watering.	– Visit different water sources and name them. – Discuss and brainstorm different types of water. – Make the rain cather using local materials through engineering process. – Discuss and brainstorm the uses of water in daily life. – Practice how to wash a handkerchief using clean water and soap. – Practice how to clean raw food like carrot, sweet potatoes, passion fruits.

Links to other subjects: *Social and Religious Studies (good hygiene and cleanliness), English (learning new vocabulary), Mathematics (counting), Creative Arts (drawing or coloring pictures)*

Generic competences:

- **Critical thinking:** Developed through identifying and classifying different sources and types of water, and explaining the importance of water in everyday life.
- **Communication (Literacy and Numeracy):** Developed through discussing sources, types, and uses of water, and presenting ideas during classroom activities.
- **Co-operation, interpersonal relations and life skills:** Developed through working together during practical activities such as washing clothes and cleaning food, while sharing water responsibly.

Basic values:

- **Dignity and integrity:** Developed through practicing good hygiene, using water responsibly, and taking care of water sources.
- **Respect for others and human rights:** Developed through sharing water fairly with others and encouraging hygienic practices within the family and community.
- **Hard work:** Developed through active participation in washing clothes, cleaning food, and maintaining personal and environmental cleanliness.

Cross-cutting issues

a) Environment and sustainability

- Help pupils understand the importance of protecting water sources (e.g., rivers, wells, tanks).
- Teach them not to throw rubbish into or near water sources.
- Encourage saving water and using it responsibly.

b) Gender education

- Show that both boys and girls are responsible for fetching water, cleaning, and using water wisely.
- Promote equal participation in class activities about water.

c) Peace and value education

- Use activities that promote cooperation when using or fetching water.
- Encourage respect and sharing when using water materials or role-playing home/school water use.

d) Inclusive education

- Ensure that all pupils, including those with disabilities, can participate in activities related to water. For example, when drawing pictures of water use, allow different forms of expression (drawing, speaking, showing with objects).

Assessment criteria: *Assess pupils' ability to name sources of water, identify clean and dirty water, state the uses of water, and show care when handling water materials.*

Materials: *Pictures or flashcard of water sources, water types and water use, bottles or cups, papers and crayons, charts/posters showing clean, dirty water, and people using water in different ways, different raw food.*

Topic area: Environment

Science and Elementary Technology:
Primary one

Unit 4: Animals

Number of periods:
10

Key unit competence: Identify domestic, wild and aquatic animals and their importance.

Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Identify domestic animals. – Name domestic animals. Explain why domestic animals are important. – Identify wild animals. – Name wild animals. – Explain why wild animals are important. Identify aquatic animals.	– Differentiate domestic, wild, and animals. – Describe the importance of each type of animal. – Match animals with their correct group. – Match animals with their correct importance.	– Show interest in learning about all kinds of animals. – Show respect and care for all animals. – Appreciate the role of animals.	Types of animals and their importance: – Domestic animals and their importance: cows, goats, rabbits, hens, dogs, cats, sheep, cows. – Wild animals and their importance: elephants, gorillas, snakes, zebra, lion, tiger, monkey.	– Draw favorite animals in their environment (where they live). – Visit a nearby farm or use pictures to observe the domestic animals and identify their importance. – Use pictures to name wild animals and explain their importance. – Use pictures to identify examples of aquatic animals and their importance.

Name aquatic animals Explain why aquatic animals are important.	– Protect animals in our homes, water, and forests.		– Aquatic animals and their importance Fish, crocodile, hippopotamus Insects: Ants, butterflies, coackroches.	– Differentiate types of animals according to where they live. – Draw and shade different domestic, wild and aquatic animals.
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Links to other subjects: *Social and Religious Studies (importance of animals), English (learning new vocabulary), Mathematics (counting animals), Creative Arts (drawing and coloring different animals)*

Generic competences:

- **Critical thinking:** Developed through identifying, classifying, and comparing domestic, wild, and aquatic animals according to where they live and explaining their importance.
- **Communication (Literacy and Numeracy):** Developed through naming different animals, discussing their characteristics and importance, and presenting findings during classroom activities.
- **Co-operation, interpersonal relations and life skills:** Developed through group discussions, observation activities, and collaborative classification of animals.

Basic values:

- **Respect for others and human rights:** Developed through showing care, kindness, and respect for animals and appreciating their importance to people and the environment.
- **Dignity and integrity:** Developed through protecting animals, avoiding cruelty, and behaving responsibly when interacting with them.
- **National and cultural identity:** Developed through appreciating Rwanda's wildlife as part of the country's natural heritage and recognizing the importance of conserving it.

Cross-cutting issues

a) Environment and sustainability

- Help learners understand that wild animals live in forests, parks, or water bodies, and that we should protect their homes (habitats).
- Teach pupils not to hurt or chase animals, and to keep the environment clean so animals can live safely.

b) Peace and values education

- Encourage respect for all living things, and show that animals should be treated
- with care, just like people.

c) Inclusive education

- Make sure every child can participate in naming animals and talking about them, including learners with different abilities or backgrounds.
- Use visuals, gestures, or real-life examples to support learners who may need extra help understanding animal names or sounds.

Assessment criteria: *Assess pupils' ability to name different animals, classify them as domestic, wild, or aquatic, explain their importance correctly.*

Materials: *Flashcards or pictures of domestic animals, wild animals, and aquatic animals, charts/posters showing groups of animals, animals and what they give us, drawing paper and crayons.*

Topic area: Environment

Science and Elementary Technology:
Primary one

Unit 5: Plants

Number of periods: 8

Key unit competence: Name plants that people grow and plants that grow on their own.

Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– List the plants found around the school. – Name plants that people grow in the community (cultivated plants). – Name plants that grow on their own in the community (non-cultivated plants).	– Group plants in cultivated plants and non-cultivated plants. – Plant new plants at school or at home. – Match plants with their correct group.	– Show interest in learning about different types of plants found at school or at home. – Take care of plants by watering them and avoiding damaging them.	Types of plants: – Cultivated plants Beans, maize, sorghum, banana plants, , tea, coffee, pear, avocado – Non-cultivated plants Eucalyptus, goat apple, Aloe vera, Euphorbia, blackjack.	– Draw and colour different plants. – Do guided research in different books from the school library to identify different types of plants that grow in their community and at school. – Visit the school garden or surroundings to observe different types of plants, uproot some of different types and bring them to school. – Discuss different plants and group them based

				on whether they are cultivated or not cultivated. – Plant some plants in the school garden and water them.
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Links to other subjects: *Social and Religious Studies (care for the environment), English (learning new vocabulary), Mathematics (grouping plants), Creative Arts (drawing and coloring different plants)*

Generic competences:

- **Critical thinking:** Developed through observing, identifying, comparing, and classifying plants into cultivated and non-cultivated groups.
- **Research and problem solving:** Developed through guided research in books and observation of plants in the school and community to identify different types of plants.
- **Co-operation, interpersonal relations and life skills:** Developed through working together during plant observation, classification, planting, and watering activities.

Basic values:

- **Dignity and integrity:** Developed through taking responsibility for planting, watering, and protecting plants at school and at home.
- **Hard work:** Developed through active participation in planting and caring for plants and maintaining the school garden.
- **National and cultural identity:** Developed through appreciating the importance of plants commonly grown in Rwanda and recognizing their contribution to people's livelihoods and the environment.

Cross-cutting issues

a) Environment and sustainability

- Help pupils understand the importance of taking care of plants and not damaging natural vegetation.
- Guide learners to observe how plants make the environment beautiful and why we should not cut trees unnecessarily.

- Encourage them to plant and water trees around the school or at home as part of good environmental behavior.

b) Peace and value education

- While learners work in groups during the outdoor plant observation activity, remind them to respect each other's ideas, share observations, and take turns speaking.
- Emphasize cooperation, patience, and care while working together and when handling living things.

c) Inclusive education

- Ensure that all pupils can participate in the outdoor observation activity, including those with physical or learning difficulties
- The teacher can assign study partners or adjust the task for learners who may not be able to walk far by showing them plants closer to the classroom or using illustrations.

Assessment criteria: *Assess pupils' ability to name plants and group them correctly according to their groups (cultivated plants, non-cultivated plants) from their community and the school area.*

Materials: *Pictures or flashcards of cultivated plants, non-cultivated plants, drawing paper and crayons, charts or posters showing plant types and their uses, plant environments, different textbooks.*

Topic area: Human body

Science and Elementary Technology:
Primary one

Unit 6: The main parts of the human body

Number of periods: 18

Key unit competence: Name the main parts of the human body, describe what they do, and keep them clean.

Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Name the main parts of the human body. – Name the main parts of the head. – Point out what different body parts do. – Talk about how to take care of the body.	– Point to the main parts of the human body. – Wash different body parts properly. – Do simple exercises.	– Recognize and respect the body and what it does. – Keep clean by washing body parts regularly. – Take routine of brushing teeth always.	– Main parts of human body: head, trunk, arms and legs. – Main parts of the head: eyes, ears, nose, tongue, mouth and teeth. – Taking care of the human body: • Washing our body • Washing Hands	– Draw human body poster and label it. – Observe pictures and touch the main human body parts. – Observe picture and touch the main parts of the head and identify their uses. – Discuss how human body parts can be cleaned and practice washing them. – Practice cleaning some parts of the body: Washing hands, brushing the teeth, cleaning nose, ...

		– Make a habit of doing exercises to stay strong and healthy.	• Brushing the teeth • Cleaning the Nose • Doing sports	– Observe pictures and discuss the importance of physical exercise. – Do physical exercise
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Links to other subjects: *Social and Religious Studies (cleanliness), English (learning vocabulary), Mathematics (counting body parts), Creative Arts (drawing and coloring parts of the body), Physical Education and Sport (doing physical activities)*

Generic competences:

Communication (Literacy and Numeracy): Developed through naming body parts, discussing their functions, and explaining ways of keeping the body clean and healthy.

- **Critical thinking:** Developed through identifying different body parts, matching them with their functions, and explaining the importance of good hygiene and physical exercise.
- **Co-operation, interpersonal relations and life skills:** Developed through participating in group discussions, body care demonstrations, and physical exercises while encouraging one another to adopt healthy habits

Basic values:

- **Dignity and integrity:** Developed through maintaining personal hygiene, respecting one's body, and practicing healthy habits such as washing, brushing teeth, and exercising regularly.
- **Self-reliance:** Developed through independently practicing daily body care routines, including handwashing, brushing teeth, and keeping the body clean.
- **Hard work:** Developed through consistently practicing healthy routines and participating actively in physical exercises to maintain good health.

Cross cutting Issues

a) Gender education

- Ensure that both boys and girls are equally involved in classroom participation (e.g., naming and pointing to body parts).
- Emphasize that all human bodies are important, regardless of gender.
- Avoid reinforcing gender stereotypes (e.g., “boys are stronger,” “girls are more careful”). Instead, promote equal abilities and respect.

b) Inclusive education

- Use pictures or classroom examples that represent pupils with different body parts and abilities.
- Adapt activities to support pupils with disabilities, such as using tactile learning for visually impaired pupils.
- Promote a classroom culture of respect and support for all pupils, regardless of learning differences.

c) Comprehensive sexuality education (CSE)

- Introduce the importance of personal hygiene and respecting private body parts using age-appropriate language.
- Teach that some parts of the body are private, and no one should touch them without permission.
- Reinforce the message that if a pupil feels sick, they should tell a trusted adult.

Assessment criteria: *Assess pupils’ ability to name body parts, show ways to take care of them and identify the uses of parts of head correctly.*

Materials: *Charts or posters showing main parts of the body, parts of the head (eyes, ears, nose, tongue, mouth, teeth), body care practice, toothbrush, soap, towel, handwashing basin, mirror etc.*

6.3. Science and Elementary Technology for Primary Two (P2)

6.3.1. General competences

At the end of P2, pupil should achieve the following main competences:

- Identify natural things and artificial (man-made) things and show interest and care in protecting them.
- Make toys, teaching and learning materials and other simple things.
- Name the main parts of a computer and use the computer correctly and safely.
- Name the main parts of a plant, list their importance, and group plants by their use.
- Sort different types of waste and keep the environment clean.
- Identify and use sources of light and heat safely and explain their importance in daily life.

6.3.2. Table units for Primary 2

Topic area: Environment				
Science and Elementary Technology: Primary two		Unit 1: Natural and artificial things		Number of periods: 8
Key unit competence: Identify natural and artificial things and demonstrate care for protecting natural things.				
Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Define natural things. – Point out natural things. – Recall what artificial things are. – Identify natural and artificial things. – Name some artificial things.	– Explain the difference between natural things and artificial things. – Match different things to their group.	– Show interest in knowing the difference between natural and artificial things. – Take care of natural things and keep them safe from harm. – Take care of artificial (man-made) things	– Natural things: Animals, rivers, mountains, trees, stones, sand, water, and others. – Artificial things (man-made things): Cars, clothes, chairs, bicycles, bottles, boats and others.	– Observe picture or move around the school to discuss and find out various natural things and artificial (man-made) things. – Sort out various natural things and man-made things. – Discuss and brainstorm to find more examples of natural things and artificial (man-made) things

Links to other subjects: *Social and Religious Studies (teaching respect and care for nature), English (learning new vocabulary), Mathematics (counting objects), Creative Arts (drawing and coloring natural things and artificial things)*

Generic Competences:

- **Critical Thinking:** In this unit, pupils observe different objects and distinguish between natural things and artificial (man-made) things by explaining why each object belongs to a particular group.
- **Communication (Literacy and Numeracy):** *Pupils discuss their observations in groups, name natural and artificial things, explain the differences between them, and present their findings to the*

Basic values:

- **Respect for Others and Human Rights:** Pupils develop respect for natural resources and artificial objects by handling them carefully and appreciating their importance in everyday life.
- **Dignity and Integrity:** Pupils demonstrate responsibility by taking care of natural resources such as trees and water, as well as man-made materials like chairs, bottles, and books.
- **Hard Work:** Pupils actively participate in observing, identifying, sorting, and classifying different objects while completing classroom activities responsibly.

Crosscutting issues:

Environment and sustainability

- Help pupils understand that natural things (like trees, soil, water) come from nature and must be protected.
- Discuss how careless use of artificial things (like plastics) can harm the environment.
- Encourage pupils to care for natural things in their surroundings (e.g., not cutting trees, not littering).

Standardization culture

- Teach pupils to recognize good quality artificial products (e.g., safe school materials).
- Discuss how using standard items like proper school uniforms and safe tools helps in learning and safety.
- Highlight the importance of using safe and approved materials for daily use.

Inclusive education

- Involve all pupils equally, including those with learning difficulties, by using simple and clear language.
- Provide opportunities for every child to share their observations of natural and artificial things.
- Use group activities to ensure participation of shy or less confident pupils.

Assessment criteria: *Assess if pupils can name, group, and talk about natural and artificial (man-made things) clearly*

Materials: *Flashcards or pictures of natural things and artificial (man-made), charts/posters, tones, leaves, small bottles, toy cars, clothes, drawing paper and crayons.*

Topic Area: Environment**Science and Elementary Technology,
Primary two****Unit 2: Making toys and classroom
learning materials****Number of periods: 12****Key unit competence:** Make simple toys and teaching and learning materials.

Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Identify the steps for making a bird toy using paper. – Identify the steps of making a toy car using dry sticks, straws, or wires. – Identify the steps for making airplane toys using empty plastic bottles	– Cut and fold paper to make a bird toy. – Use sticks, straws, or wires to make a toy car. – Use a plastic bottle to make an airplane toy. – Use a box to make a toy wall clock.	– Be careful when using sharp things. – Be curious and try to make your own toys instead of buying them. – Take care of toys, Keep them safe and clean. – Show creativity while making toys.	– Making a toy bird out of paper. – Making a toy car out of dry sorghum sticks, straws or wires. – Making a toy airplane out of an empty plastic bottle. – Making a wall clock toy out of a box.	– Make bird toy by following the engineering process . – Fly the toy bird made in the school compound. – Making a toy car out of dry sorghum sticks using straws or wires by following given steps. – Make airplane and car toys using sorghum sticks, tree sticks and papers by following the engineering process.

– Identify the steps of making a wall clock toy using a box. – Recognize different toys. – Identify the materials used to make toys. – Explain the uses of toys to children.	– Cut, fold, and join materials to make toys. – Use hands and tools safely to create toys. – Follow simple steps to complete a task. – Make more different toys out of locally available resources.	– Work neatly and carefully. – Share tools and materials with friends. – Use local and recycled materials responsibly. – Be playful and have fun with others – Be careful while playing with different toys.		– Drive the toy car and fly a toy airplane in the school compound. – Follow the steps and make a wall clock toy using boxes. – Discuss to discover more toys that can be made out of paper, dry sorghum sticks, straw or wire and box. – Make more toys and learning materials out of paper, dry sorghum sticks, straw or wire and box.
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Links to other subjects: *Creative Art (drawing and coloring), Mathematics (measuring lengths), English (learning new vocabulary),*

Generic Competences

Creativity and Innovation

- Learners design and make different toys and learning materials using locally available and recycled materials.

Critical Thinking

- Learners follow steps, select appropriate materials, and solve simple problems while making toys.

Co-operation, Interpersonal Relations and Life Skills

- Learners work together, share tools and materials, and support one another during practical activities.

Basic Values

Hard Work

- Learners make toys carefully by following the correct procedures and completing tasks responsibly.

Self-Reliance

- Learners develop confidence by making their own toys instead of depending on buying them.

Respect for Others and Human Rights

- Learners share materials, work cooperatively, and respect others while carrying out classroom activities.

Cross-cutting Issues

Gender Education

- Promote equal participation of both boys and girls in making toys and learning materials.
- Avoid assigning tasks like cutting, drawing, or designing based on gender.
- Use inclusive language: “All of us can make toys and learning materials.”
- Encourage mixed-gender teamwork to develop cooperation and break down stereotypes.

Environment and sustainability

- Encourage pupils to use recycled and eco-friendly materials when making toys and learning aids.
- Guide pupils to collect safe and clean waste materials like boxes, bottles, or papers for making toys.
- Explain how recycling helps reduce waste and protect the environment.
- Lead a class discussion: “How can we make toys without harming nature?”

Inclusive education

- Ensure that all pupils, including those with special needs, can participate in making toys and learning materials.
- Provide adapted materials (e.g., larger scissors, textured items) for pupils with physical or visual impairments
- Encourage group work and peer support.
- Recognize different talents, some pupils may be better at designing, others at assembling

Assessment criteria: *Assess pupils’ ability to choose the right materials, follow steps to make any toy, talk about how they made it, and play with it confidently.*

Materials: *Pair of scissors, razor blade, papers, boxes, sorghum sticks, tree sticks, glue, nails, rubber, hard paper, empty plastic bottles, cardbox, Crayons or markers*

Topic Area: ICT

Science and Elementary
Technology
Primary two

Unit 3: Computer my Friend

Number of periods: 14

Key unit competence: Name the main parts of a computer and use the computer correctly and safely.

Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Name the outer parts of a computer. – Mention the role of each part of a computer. – Identify ways to take care of computers. – Describe ways to take care of a computer.	– Point to and name the outer parts of a computer. – Match the parts of a computer to their uses. – Show the difference between a laptop computer and desktop computer.	– Take care of all computer parts. – Show interest in using the computer. – Sit properly when using a computer to keep your body safe. – Keep the computer clean and use it gently	– The parts of a laptop computer Screen, keyboard, touchpad, adaptor – The parts of a desktop computer: Screen, keyboard, mouse, computer case, power cable. – Uses of the parts of a computer: – Taking care of a computer	– Visit a computer lab to observe and identify different parts of a computer. – Visit a computer lab to identify the difference between a laptop computer and a desktop computer. – Label diagrams of a desktop and laptop computer with their correct part names. Discuss and brainstorm about external parts of the computer and their functions.

– Identify the proper sitting posture while using a computer – Name the dangers of poor sitting while using a computer.	– Use the computer in a safe way to avoid harm. – Follow rules when using the computer and turn it off the right way. – Sit well when using a computer.	– Take care of health by sitting well when using a computer.	– Sitting posture when using a computer • Proper sitting postures • Improper sitting postures – Dangers of poor sitting posture while using a computer	– Discuss and brainstorm about the ways of taking care of a computer. – Make exercise of appropriate sitting while using a computer. – Discuss dangers to health from inappropriate sitting while using a computer.
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Links to other subjects: *Mathematics (Counting), English (Learning vocabulary), Physical Education (correct sitting posture), Social and Religious Studies (good habits when using technology), Creative Arts (Drawing or labeling computer parts)*

Generic competences

- **Communication (Literacy and Numeracy):** Learners name computer parts, discuss their functions, and explain how to use and care for a computer correctly.
- **Co-operation, Interpersonal Relations and Life Skills:** Learners work together to identify computer parts, practice correct sitting posture, and follow computer lab rules.
- **Critical Thinking:** Learners compare desktop and laptop computers, identify their differences, and match computer parts with their uses.

Basic values

Responsibility and Integrity

- Learners use computers carefully, follow safety rules, and take care of computer equipment.

Respect for Others and Human Rights

- Learners respect classmates by sharing computers, taking turns, and using ICT resources responsibly.

Hard Work

- Learners actively practice identifying computer parts, maintaining proper posture, and caring for computers.

Crosscutting issues

- **Gender Equality:** Make sure that pupils understand that computers are used both girls and boys and show examples of both men and women using computers.
- **Environment and sustainability:** Tell pupils that computers must be used responsibly so that they do not consume too much electricity. Explain why we should not throw old computers in nature.
- **Financial education:** Explain to pupils that computers are bought at different costs and that buyers must buy depending on what they want to use a computer. Tell them the prices of some computers.

Assessment criteria: *Assess if the pupils' ability to name different parts of a computer, identify their uses, show good sitting posture, and explain how to take care of a computer efficiently.*

Materials: *Different types of computer, markers and papers, videos, charts or pictures, flashcards with names of parts of a computer, etc.*

Topic Area: Our environment

Science and Elementary Technology,
Primary Two

Unit 4: Plants

Number of periods: 8

Key Unit Competence: Name the main parts of a plant and explain the importance of different plants.

Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Name the main parts of a plant. – List the difference between the plants according to their uses. – Identify the importance of plants.	– Draw a plant with all the main part. – Sort plants according to their uses. – Label the parts of a plant on a picture/ image and real plant sample.	– Be thankful for how plants help us. – Take care of plants. – Respect nature by avoiding to harm the plants.	– Main parts of the plant and their functions Roots, stem, branches, leaves, flowers, fruits and grains. – Importance of plants: Give food, fresh air, firewood and timber, medicines, money, decoration and protect the soil against erosion.	– Visit the school compound to observe the different parts of plants. – Brainstorm and discuss to identify parts of the plant and functions of every part. – Draw a plant with the main parts and color each part of it. – Discuss and identify different importance of plants. – Discuss and group plants according to their uses. – Draw and shade plants and its parts.

Links to other subjects: *Creative Arts(drawing), Mathematics (Counting), English(Learning new vocabulary), Social and Religious Studies (role of plants)*

Generic Competences

- **Communication (Literacy and Numeracy):** Pupils name the main parts of a plant, describe their functions, and explain the importance of plants during class discussions and presentations.
- **Critical Thinking:** Pupils compare different plants, identify their uses, and classify them according to their importance in everyday life.
- **Co-operation, Interpersonal Relations and Life Skills:** Pupils work together to observe plants, identify their parts, and complete group activities such as drawing and classifying plants.

Basic Values

- **Dignity and Integrity:** Pupils take care of plants and demonstrate responsible behavior by protecting them from damage.
- **Respect for Others and Human Rights:** Pupils respect nature and appreciate the role plants play in supporting the wellbeing of all living things.
- **Hard Work:** Pupils actively participate in observing, drawing, labeling, and grouping plants according to their uses.

Cross-cutting Issues

Environment and sustainability

- Emphasize how plants help maintain environmental balance (e.g., they purify air, prevent soil erosion, and provide shade and habitat for animals).
- Guide pupils to discuss what would happen if there were no plants, highlighting the importance of planting trees and caring for vegetation.
- Encourage pupils to engage in small actions, such as not cutting trees unnecessarily or helping water school garden plants.

Financial education

- Highlight how some plants (e.g., coffee, tea, fruits, and vegetables) are sources of income for families and the country.
- Ask pupils if they know anyone in their community who earns a living by growing and selling plants or plant products.
- Encourage pupils to view agriculture not only as a survival activity but also as an opportunity to generate income in the future.

Gender education

- Promote equal participation by both boys and girls when taking pupils outside the classroom or during discussions.
- Use examples of successful men and women in agriculture to inspire pupils, reinforcing that both genders can contribute equally to environmental protection and farming.

Assessment criteria: *Assess pupils' ability to identify main parts of a plant, describe their functions, and explain the importance of plants in daily life correctly.*

Materials: *Charts or posters showing parts of plant, crayons, flashcards with names of parts plant, real plant, etc.*

Topic Area: Our environment**Science and Elementary Technology:
Primary Two****Unit 5: Waste and hygiene****Key Unit Competence:** Sort different types of waste and keep the environment clean.**Learning objectives****Knowledge and understanding**

- Name the types of waste.
- Define decomposing waste.
- Define non-decomposing waste.
- Define toxic waste.
- Explain the ways of waste management.
- Identify the ways of keeping environment clean.

Skills

- Differentiate the three types of waste.
- Sort wastes into the right bins accordingly to their types.
- Clean home and school.
- Use appropriate cleaning materials to clean the waste.

Attitudes and values

- Care about the hygiene of the surroundings.
- Get used to sorting waste properly.
- Be careful not to get hurt by sharp things like broken bottles.
- Stay away from toxic waste
- Be careful while managing the wastes.

Content

- **Types of Wastes:**
 - Decomposing wastes (grasses, leaves of plants, fruits, vegetables, dead animals and papers).
 - Non-decomposing waste (metals, plastics, glasses and construction waste).
 - Toxic wastes (used batteries, dry cells and old medicines).

Learning activities

- Discuss and brainstorm different types of waste.
- Walk in the school compound to collect different types of waste and put them in the right dustbin according to their types.
- Design a public campaign poster to encourage the separation of waste.
- Discuss in groups to differentiate types of waste and ways of waste management.

	–	– Appreciate the importance of waste separation to the environment.	– Ways of waste management: • Cleaning the home and community • Separation of wastes	– Organize cleaning activity of the classroom by collecting and sweeping all wastes on the floor and dispose them in separate and appropriate dustbins.
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Links to other subjects: *Social and Religious Studies (Promoting cleanliness), Entrepreneurship (recycle waste into useful or sellable items), English (Learning new vocabulary), Mathematics (counting types of waste)*

Generic Competences

- **Critical Thinking:** Pupils classify waste into decomposing, non-decomposing, and toxic waste.
- **Communication, Literacy and Numeracy:** Pupils present their findings on waste separation and environmental cleanliness.
- **Cooperation and Interpersonal Relation:** Pupils support one another in keeping the school and home environment clean.

Basic Values

Hard Work

- Pupils actively participate in cleaning the classroom and school compound, sort waste into the correct bins, and develop responsibility for maintaining a clean environment through consistent effort.

Solidarity and Democracy

- Pupils work together during waste collection, separation, and cleaning activities, sharing responsibilities and cooperating to keep their school and community clean.

Cross-Cutting Issues

Environment and sustainability

- Teach pupils how proper waste management protects the environment.
- Guide them to sort waste (e.g., plastic vs. food waste) and organize a school clean-up.

Peace and value education

- Encourage teamwork and responsibility during cleaning activities.
- Promote respect for shared spaces and working together peacefully.

Gender Education

- Assign cleaning tasks to both boys and girls equally.
- Talk about how hygiene is everyone's responsibility, not just for one gender.

Assessment criteria: *Assess if the pupil can identify different types of waste and explain simple ways to manage them, such as separating and cleaning.*

Materials: *Posters showing waste types, flashcards of waste items, broom, dustpan, gloves, dry leaves, fruit peels, paper scraps, pieces of plastic, metal bottle tops, glass, batteries, expired medicine wrappers.*

Topic Area: Energy

**Science and Elementary Technology,
Primary Two**

Unit 6: Light and heat

Number of periods:12

Key Unit Competence: Identify sources of light and heat, use them safely, and explain their importance in daily life.

Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Name sources of light and heat. – Recall how light and darkness are related. – Recall what a thermometer is for. – Explain where shadows come from. – Explain why light is important for living things. – Give reason why too much or too little light can be bad.	– Use light and heat in a safe way. – Measure body temperature using a thermometer. – Avoid the dangers from too much light or heat. – Adjust light when reading or working.	– Be careful to stop accidents from heat. – Use just the right amount of light when reading. – Avoid looking at the sun without glasses. – Practice good habits to stay safe from light and heat.	– Sources of light: sun, torch, fire, candle, lanterns, bulbs, matchstick, lamp, fire wood and others – Darkness and shadow – The importance of light for living things – Too much light and too little light: – Dangers of light and ways of avoiding them	– Discuss in pairs to find out various sources of light. – Discuss in pairs the importance of light. – Find out appropriate behavior to avoid the dangers of light. – Make a dark room using opaque materials like clothes. – Produce a shadow using the source of light (torch light, candle light). – Measure the height of shadow of a tree in school garden and at different hours of the day (morning, midday, evening).

- Point out how to avoid problems from too little light.
- Discuss the relationship between sources of light and sources of heat
- Explain why heat is important and when it is dangerous.
- Name the tool to measure heat (thermometer).

- Practice how to use a thermometer to check body temperature.

- Measure your body temperature regularly.

- **Sources of heat:** Sun, candle, lantern, electrical bulb, match stick, lamp, firewood and others.

- Importance of heat
- Dangers of too much heat
- Instruments to measure heat
- Measuring body temperature
- Relationship between sources of light and sources of heat

- Discuss in pairs to find out the difference between darkness, shadow and light.
- Experiment: To put on different sources of light (candle, spirit lamp, electrical bulb) with the purpose of finding out the relationship between sources of light and sources of heat.
- Discuss in pairs to find out sources of light which are not sources of heat.
- Experiment to establish the difference/relationship between the light and heat using candle, spirit lamp and electric bulb.
- out the advantages and dangers of heat.
- Group discussion on how to prevent the dangers of heat.
- To investigate different intensity of heat during different sun intensity

				– Discuss in group to find out the advantages and dangers of heat. – Experiment: Use medical thermometer to measure body temperature – Measure the temperature of various objects (air, soil, water) during different hours of the day (morning, mid-day, evening) and compare the readings.
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Links to other subjects: *Physics (Energy), Mathematics (Measuring temperature), Creative Arts (Drawing or coloring pictures), English (Reading or naming sources), Social and Religious Studies (safety around fire or electricity)*

Generic Competences

• Critical Thinking

Pupils compare different sources of light and heat, identify their relationship, and distinguish between light, darkness, and shadows.

• Communication, Literacy and Numeracy

Pupils discuss their observations in pairs or groups, present their findings, and compare thermometer readings using appropriate scientific vocabulary.

• Research and Problem Solving

Pupils investigate sources of light and heat through experiments and identify safe ways of using them in everyday life.

Basic Values

• Self-Reliance

Pupils develop the ability to make safe decisions when using sources of light and heat, measure temperature correctly, and solve simple problems related to light and heat in daily life independently.

• Responsibility

Pupils use light and heat sources carefully, follow safety rules to prevent accidents, and take care of their own wellbeing and that of others when handling heat and electrical materials.

Cross-Cutting Issues

• Environment and sustainability

- Explain how the sun (a natural source of light and heat) supports plant growth and life on Earth.
- Discuss responsible energy use and avoiding waste of artificial light or heat.

• Gender education

- Ensure all pupils, both boys and girls, participate equally in experiments and discussions.
- Promote gender equality in science-related learning activities.

• Inclusive education

- Support all pupils, including those with visual or learning challenges, by using varied teaching materials like tactile objects or simplified visuals to explain light and heat concepts.

Assessment criteria: *Assess if the pupil can name sources of light and heat, explain their importance, identify dangers, and describe ways to stay safe.*

Materials: *Sun, torches, fire, candles, spirit lamp, petrol lamp, electric lamp/bulb, matchbox, gas lamp, firewood, boxes, opaque curtains, ruler, moon, stars, thermometers, pens, papers, soil, boiling water, Flashcards with names of light sources, Pictures or drawings of the sun, fire, or bulb*

6.4. Science and Elementary Technology for Primary Three (P3)

6.4.1. General competences

At the end of P3, pupil should achieve the following main competences:

- Identify, use, and care for common agricultural tools safely and correctly.
- Write small letters and capital letters in a Word Processing program using the keyboard.
- Prepare, store, and drink clean water in a safe and healthy way.
- Identify and tell the difference between animals with a backbone and those without a backbone, and to show care for animals in the environment.
- Use materials that use electricity in a safe way and describe how electricity helps us.
- Identify magnets, use them safely and know the difference between magnetic and non-magnetic objects.

6.4.2. Table units for Primary 3

Topic Area: Elementary technology				
Science and Elementary Technology: Primary Three			Unit 1: Agricultural tools	Number of periods: 12
Key unit competence: Identify agricultural tools and safely use and care for them.				
Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Identify common agricultural tools. – Name common agriculture tools. – Explain the use of each tool. – Describe how to care for and maintain farm tools.	– Draw and name different agricultural tools. – Sort tools based on their use. – Apply the correct tool for each agricultural task.	– Use agricultural tools carefully and responsibly. – Pay attention to safety when using tools. – Show concern about preventing accidents caused by misuse of tools.	– The common Agriculture tools: Hoe, spade, rake, axe, wheelbarrow, machete, watering can, trowel, milking can, spraying pump, file. – Uses of agriculture tools:(clearing tools, digging tools, watering tools, transplanting tools, sharpening tools, spraying tools, cleaning tools and safety tools).	– Observe the range of agriculture tools, name them and match them with their use. – Visit an agricultural farm to observe how to handle, and practice proper maintenance of common agricultural tools and practice them. – Role play about the use of and care for agriculture tools.

- Explain the dangers of using tools the wrong way. - State safety rules to follow when using farm tools.	- Practice proper cleaning and maintenance of tools. - Inspect tools before and after use for any damage. - Store tools in the correct place after use.	- Develop responsibility in keeping tools in good condition and safe to use.	- Maintenance of agricultural tools:(Sharpening, oiling, cleaning, repairing or replacing damaged parts of tools, keeping them in a dry place). - Storage of agriculture tools - Dangers of the misuse of agricultural tools • Precaution when using agricultural tools.	- Discuss in group about the prevention against possible dangers of misusing of agricultural tools and make a presentation. - Draw different agriculture tools. - Design and make agricultural tools using local materials through engineering process - Practice more different activities in terms of identifying, using, and caring for common agricultural tools safely.
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Links to other subjects: *Mathematics (length or weight of tools), English (learning new vocabulary), Social and Religious Studies (exploring community farming practices), Creative Arts (drawing tools), Entrepreneurship (promoting ideas for tool maintenance services)*

Generic Competences:

- **Critical thinking:** Developed through identifying, comparing, and classifying agricultural tools according to their functions, selecting suitable tools for farming activities, and explaining safe use and maintenance practices.
- **Cooperation, interpersonal relations and life skills:** Developed through collaborative farm visits, role plays, and group discussions on the safe use, maintenance, and storage of agricultural tools, promoting teamwork and communication skills.
- **Research and problem solving:** Developed through observing farming practices, investigating tool maintenance methods, identifying risks of misuse, and suggesting practical safety measures.

Basic values:

- **Responsibility (Dignity and Integrity):** Developed through using agricultural tools safely, cleaning and maintaining them properly, and storing them correctly after use.
- **Hard work:** Developed through active participation in practical farming activities, tool maintenance, and demonstrations, promoting diligence and a positive attitude toward work.
- **Self-reliance:** Developed through acquiring practical skills in selecting, using, maintaining, and caring for agricultural tools to perform simple farming activities independently.

Cross cutting Issues

- **Peace and values:** When using digging and clearing agricultural tools, attention is needed in order to prevent accidents to each other. E.g: Pupils work in pairs to dig a small plot, taking turns and supporting each other rather than competing or fighting over tools
- **Gender equality:** Encourage both boys and girls to participate equally in using and managing agricultural tools. E.g: Let girls use heavy tools like spades and boys use lighter tools like watering cans, showing that all pupils can learn and perform equally regardless of gender.
- **Inclusiveness and equity:** Ensure that all pupils, including those with disabilities, can participate in practical work with tools e.g: by adapting tools or offering support.

- **Environment and sustainability:** Teach pupils how to use tools in ways that do not damage the soil or natural resources. E.g: While learning about a hoe, pupils discuss how improper tilling can lead to soil erosion, and how to prevent it by using conservation techniques like contour farming.

Assessment criteria: *Assess the pupil's ability to identify, name, and explain the uses and maintenance of common agricultural tools correctly and safely.*

Materials: *Hoe, Spade, Rake, Axe, Wheelbarrow, Machete, Watering can, Trowel, Milking can, Spraying pump, File, Flashcards or charts showing names and pictures of tools, posters or diagrams showing parts of tools and their uses, pens/pencils, Tool rack, Gloves, aprons, or protective gear, Soil or sand for practice, etc.*

Topic Area: ICT

Science and Elementary Technology:

Primary Three

Unit 2: Writing with a computer

Number of periods: 10

Key unit competence: Write small letters and capital letters in a Word Processing program using the keyboard.

Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Identify the main parts of the keyboard and touchpad. – Show how to open and save a Word document. – Identify the Word processing tools used to change the text font.	– Open a word processor, create a new document, and save it with a name. – Type text properly using the correct fingers and correct hands position.	– Show interest in using a computer to write. – Enjoy typing and formatting text on a computer. – Be curious to explore the toolbar and text area in a word processor.	– The Keyboard and Touchpad • Parts of the computer keyboard • Finger placement on a keyboard. • Touchpad – Opening and saving a document • Opening a new document • Saving a word document • Opening an existing document	– Discuss parts of keyboard and touchpad. – Practice on how to place fingers on the keyboard. – Practice on how to use a touchpad. – Practice using arrow keys to move the cursor downward, upward and in different ways. – Practical exercise on opening and saving a file.

- Explain how to bold, italicize, and underline a text - Explain how to change letter case using word processing tools.	- Create and name a new word document. - Save documents to a specific folder or location. - Type text accurately in both uppercase and lowercase using keyboard - Apply formatting tools in word processors to adjust letter case.	- Enjoy learning with ICT tools like computers. - Enjoy typing text using the correct hands and correct fingers. - Show responsibility by saving work regularly to avoid loss.	- Writing capital and small letters • Writing capital and small letters using the keyboard • Writing capital and small letters using word tool	- Practical exercise on writing capital and small letters using the keyboard in word document.
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Links to other subjects: *English (learning new vocabulary), Mathematics (using numbers and symbols on the keyboard), Entrepreneurship (learning how to use a keyboard for typing documents in business)*

Generic Competences:

- **Critical thinking:** Developed through identifying keyboard keys and word processing tools, distinguishing between uppercase and lowercase letters, and selecting suitable formatting tools to create well-presented documents.
- **Research and problem solving:** Developed through exploring the keyboard, touchpad, and word processor using practical activities, identifying tool functions, and solving simple typing and document-saving challenges.
- **Communication (Literacy and Numeracy):** Developed through typing words, sentences, and simple documents accurately using correct letter cases and formatting tools to communicate ideas through digital text.

Basic values:

- **Responsibility (Dignity and Integrity):** Developed through using computers carefully, saving work regularly, and handling ICT equipment properly.
- **Hard work:** Developed through practicing keyboard skills, improving typing accuracy, and completing practical tasks with commitment and perseverance.
- **Self-reliance:** Developed through gaining confidence and skills to independently create, edit, save, and retrieve simple documents using a word processor.

Cross cutting Issues

- **Environment and sustainability:** Provide short sentences that are about protecting the environment and make pupils write those sentences.
- **Inclusive Education:** Encourage participation from all Pupils and create a supportive classroom where every learner feels valued and confident to share their ideas through writing, regardless of ability or background.

Assessment criteria: *Assess the pupil's ability to identify parts of the keyboard and touchpad, demonstrate correct finger placement, and open, type, save, and reopen a word document correctly.*

Materials: *Computers or laptops with keyboards and touchpads, XO Laptop, Projector, Charts showing parts of the keyboard, posters of finger placement for typing, word processor.*

Topic Area: Our environment

Science and Elementary Technology: Primary three

Unit 3: Drinking Water

Number of periods: 10

Key unit competence: Prepare, store, and drink clean water in a safe and healthy way.

Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Identify drinking water and dirty water. – Explain how to make safe drinking water. – Describe how to store clean drinking water properly.	– Clean the containers used to store water. – Boil water to make it safe for drinking. – Use water treatment (like Sur'Eau) to kill germs. – Wash and maintain the filter regularly – Store boiled water in clean, covered containers.	– Always drink clean or boiled water. – Take care to keep water clean and safe for health. – Be aware that boiled water should be stored in clean, covered containers. – Appreciate the importance of safe drinking water for health.	– Preparation of drinking water by boiling. – Preparation of drinking water by filtering – Preparation of drinking water by using disinfectant "SUR'EAU"	– To boil water and filter it. – In groups, discuss the advantages and disadvantages of each method (boiling, filtering, SUR'EAU) and present findings to the class. – Visit a nearby water source or school kitchen, observe how water is prepared and record the steps, tools used, and hygiene practices. Back in class and share reports and suggest improvements.

	– Match the tool to its use.	– Be patient in filtering water properly. – Appreciate the role of SUR'EAU in preventing waterborne diseases.		– To design and make a solution (filter) to keep the water clean of dirt and dust through engineering process. – Practice on preparing drinking water using a filter. – Prepare drinking water using SUR'EAU water treatment. – Practice on cleaning materials/utensils for storing water. – Practice the proper storing of water.
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Links to other subjects: *Biology (small insects), Social and Religious Studies (community access to clean water), Mathematics (Measuring amounts of water to be boiled, filtered, or disinfected), Entrepreneurship (value of safe water as a service to the community), English (learning new vocabulary)*

Generic Competences:

- **Critical thinking:** Developed through comparing methods of preparing safe drinking water (boiling, filtering, and using SUR'EAU), analyzing their advantages and disadvantages, and selecting appropriate methods for different situations.
- **Research and problem solving:** Developed through investigating water preparation practices, identifying sources of contamination, and applying practical methods to make water safe for drinking and storage.
- **Communication (Literacy and Numeracy):** Developed through group discussions, presenting findings on water treatment methods, explaining procedures using scientific vocabulary, and recording observations from practical activities.

Basic values:

- **Responsibility (Dignity and Integrity):** Developed through preparing, storing, and using safe drinking water hygienically to protect personal and community health.
- **Hard work:** Developed through active participation in practical activities such as boiling, filtering, disinfecting, and safely storing drinking water.
- **Self-reliance:** Developed through acquiring practical skills to independently prepare, treat, and store safe drinking water using locally available methods and materials.

Cross cutting Issues

- **Gender:** When boys and girls work together in the following activities: to boil water and to disinfect water using chemicals e.g. Sur'eau.
- **Environment sustainability:** For example, when pupils avoid dirt in their place of work as they boil or disinfect drinking water.
- **Peace and values:** Tell pupils to work together peacefully and cooperate.
- **Financial education:** When pupils keep responsibly the materials they use to boil or disinfect water for drinking.
- **Standardization culture:** For example, when pupils check whether Sur'eau they use to disinfect water is not expired

Assessment criteria: *Assess the pupil's ability to demonstrate and explain how to prepare safe drinking water by boiling, filtering, and using disinfectants like SUR'EAU correctly and hygienically.*

Materials: *Clean water, cooking pot or kettle, heat source (charcoal stove, gas cooker, or electric heater), cups or bottles, dirty or unclean water, clean cloth or cotton, sand and gravel, plastic bottle, container, bottle of SUR'EAU, measuring cap or spoon, clean water, charts or posters showing steps of water treatment, pictures or flashcards showing unsafe and safe water, soap and water, gloves or tongs, towel or tray for drying utensils.*

Topic Area: Environment

Science and Elementary Technology:
Primary three

Unit 4: Animals

Number of periods: 8

Key unit competence: Identify and tell the difference between vertebrate and invertebrate animals, and demonstrate care for animals in the environment.

Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– List the animals that live around us. – Identifying vertebrates and invertebrates animals. – Name animals with a backbone (vertebrates). – Name animals without a backbone (invertebrates).	– Differentiate animals with a backbone and those without a backbone. – Classify animals into their correct groups.	– Show interest in learning about animals around us. – Care for and protect animals in the environment.	– Vertebrate animals: Cow, hen, lizard, rabbit, sheep, goat, dog, pig, duck, bat, rat, fish, snake, mole, toad ... – Invertebrate animals: Housefly, butterfly, bee, mosquito, maggot, earthworm, black ant.	– Visit a place where animals are kept to observe and identify vertebrate animals found in that place. – List down examples of Vertebrates and Invertebrates. – Draw different pictures of animals and use a string or colour to show the backbone. – Work in groups to look at pictures of animals and decide which are vertebrates and which

				are invertebrates in our environment. – Carry out activities to identify and compare animals with a backbone and those without a backbone, and demonstrate ways to care for animals in the environment.
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Links to other subjects: *Biology (vertebrates and invertebrates), English (learning new vocabulary), Mathematics (counting and grouping animals), Creative Arts (drawing or modeling animals using clay or paper), Entrepreneurship (discussing how some animals generate income)*

Generic Competences:

- **Critical thinking:** Developed through observing, comparing, and classifying animals based on the presence or absence of a backbone, and identifying similarities and differences between vertebrates and invertebrates.
- **Research and problem solving:** Developed through investigating animals in the surroundings using field visits and picture observations, classifying them correctly, and suggesting ways to care for and protect animals.
- **Communication (Literacy and Numeracy):** Developed through group discussions, classifying animals, presenting findings, and using appropriate scientific vocabulary to describe vertebrates and invertebrates

Basic values:

- **Respect for others and human rights:** Developed through showing respect for living things, treating animals humanely, and recognizing the importance of protecting wildlife and domestic animals.
- **Responsibility (Dignity and Integrity):** Developed through caring for animals, protecting their habitats, and avoiding activities that may harm them.
- **National and cultural identity:** Developed through appreciating the diversity of animals in Rwanda and understanding the importance of conserving natural resources and biodiversity.

Cross cutting Issues

- **Environmental sustainability:** This will be achieved after pupils have well understood the different types of animals and how they can be protected.
- **Standardization culture:** How different societies view and value animals. Indigenous practices in animal management and conservation. For example: Teacher can explain cultural significance of animals.

Assessment criteria: *Assess the pupil's ability to correctly identify and classify animals as vertebrates or invertebrates and explain the main differences between the two groups correctly.*

Materials: *Cow, sheep, goat, rabbit, pig, duck, bat, rat, hen, lizard, fish, snake, housefly, butterfly, bee, mosquito, mole rat, pictures, photos of animals that cannot be easily obtained, flashcards with animal names and pictures,*

Topic Area: Energy

Science and Elementary Technology:
Primary three

Unit 5: Electricity

Number of periods: 14

Key unit competence: Identify and use devices that use electricity in a safe way, and state how electricity helps us.

Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Define electricity. – Identify different materials that use electricity to do work. – List the different ways electricity is used. – Name things at home and school that use electricity. – List some dangers that	– Use electrical items safely. – Practice ways to stay safe when using electricity. – Demonstrate safe ways to handle electrical devices. – Report damaged or unsafe electrical situations.	– Be careful when using things that use electricity. – Respect safety rules to stay safe. – Practice ways to stay safe when using electricity	– Meaning of electricity and its uses: • Meaning of electricity • Uses of electricity – Materials that use electricity – Dangers from electricity – Preventing the dangers of electricity	– To connect wires to different objects including batteries to see if the bulb light up. – Group discussion to find out the uses of electricity and electronic devices and how bulb works. – Individual practice on connecting, disconnecting, turning on/off, switching on/off various materials that use electricity.

are caused by electricity. – Explain how to stay safe from electric shocks, burns, and fire.	–			– Group discussion to find out the dangers from electricity (poor handling and use of electricity) – Discuss to find out the ways to prevent dangers of electricity. – Discuss and practice using electrical materials and learn how electricity helps the community. – Draw and shade materials that use electricity to do work.
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Links to other subjects: *Physics (energy), Mathematics (understanding numbers and symbols seen on devices), English (learning and using new vocabulary), Social and Religious studies (exploring different sources of electricity), Entrepreneurship (identifying electric-powered business tools),*

Generic Competencies:

- **Critical thinking:** Developed through identifying electrical devices, comparing their uses, recognizing electrical hazards, and selecting appropriate safety measures.
- **Research and problem solving:** Developed through investigating electricity use at home and school, identifying unsafe practices, and suggesting ways to prevent electrical accidents.
- **Communication (Literacy and Numeracy):** Developed through discussing electricity uses and dangers, presenting safety rules, and using appropriate science and ICT vocabulary related to electricity.

Basic values:

- **Responsibility (Dignity and Integrity):** Developed through using electrical appliances safely, reporting damaged equipment, and following safety rules at home and school.
- **Self-reliance:** Developed through gaining confidence in operating simple electrical devices safely and applying basic electrical safety measures independently.
- **Hard work:** Developed through active participation in practical activities, discussions, and demonstrations to build knowledge and safe working habits.

Cross cutting Issues

- **Environment sustainability:** Understands the importance of conserving and protecting electrical equipment.
- **Financial education:** The use of electricity in different activities like cooking, hair dressing salon.
- **Standardization culture:** When using energy generating and electrical appliances of a standard quality, that can't cause accidents.
- **Gender:** Girls and boys work together in various activities related to electricity.

Assessment criteria: *Assess the pupil's ability to explain the meaning of electricity, identify materials that use electricity, describe its common uses, recognize potential dangers, and suggest basic safety measures to prevent those dangers correctly.*

Materials: *Electronic devices: computer, telephone, flat Iron, refrigerator, torch, electric cooker, hot air conditioner, television set, radio, charts or posters showing sources of electricity (batteries, plugs, solar panels), electrical appliances (radio, bulb, TV, fan, iron, etc.), and dangers and safety signs related to electricity, batteries or power banks, electric socket*

Topic Area: Energy

Science and Elementary Technology:

Primary three

Unit 6: Magnets

Number of periods: 12

Key unit competence: Identify magnets, use them safely and state the difference between magnetic and non-magnetic objects.

Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Define a magnet. – List objects that are magnetic and objects that are non-magnetic. – Describe the difference between magnetic and non-magnetic objects.	– Differentiate magnet and other objects. – Demonstrate how a magnet attracts certain materials. – See and feel how magnetic force works.	– Appreciate how magnets are useful. – Be careful when using magnets.	– The meaning of a magnet – Magnetic and non-magnetic objects: – Characteristics of a magnet – Uses of magnets	– Designing and marking a solution using magnet to keep doors and windows open during the wind blow (Engineering process). – Bringing magnets closer to different objects and materials made from iron, copper, glasses, wood, plastics, with an objective of demonstrating and discovering magnetic objects and non-magnetic objects.

Describe what a magnet can do. – List the uses of a magnet.	– Sort objects into magnetic and non-magnetic	– Show curiosity about how magnets work.		– Placing a paper, glass, wood, or plastics in the middle of the magnet and other magnetic objects. – Designing and making a solution using magnet to keep doors and windows open during the wind blow (Engineering process). – Bringing closer to each other two or three magnets with an objective of observing and discovering what happens. – Group discussions on the dangers of magnetic forces on electronic objects and other different materials.
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Links to other subjects: *Physics (electricity), Mathematics (measuring distance where magnetic force is effective), English (learning new vocabulary), Social and religious Studies (discussing uses of magnets in homes or workplaces), Creative Arts (drawing or modeling magnets and objects), Entrepreneurship (business tools (magnetic locks, electronics).)*

Generic Competences:

- **Critical thinking:** Developed through investigating and comparing materials, classifying them as magnetic or non-magnetic, and explaining why some objects are attracted to magnets.
- **Research and problem solving:** Developed through conducting simple magnet experiments, observing magnetic forces, identifying uses of magnets, and solving classification tasks through investigation.
- **Communication (Literacy and Numeracy):** Developed through discussing experimental observations, describing magnet properties and uses using scientific vocabulary, and presenting findings.

Basic values:

- **Responsibility (Dignity and Integrity):** Developed through handling magnets carefully, protecting electronic devices from magnetic damage, and following safety instructions during practical activities.
- **Self-reliance:** Developed through independently investigating magnetic and non-magnetic materials, classifying objects correctly, and applying knowledge in daily life situations.

- **Hard work:** Developed through active participation in practical investigations, careful observation of results, and completion of classification and experimentation tasks with commitment.

Cross cutting Issues

- **Gender:** Girls and boys work together in various activities related to magnetic forces.
- **Financial education:** Pupils keep well magnetic and non-magnetic items used in all activities related to learning and understanding magnetism.
- **Environment sustainability:** Pupils leave their working place clean and safe as they found it.

Assessment criteria: *Assess the pupil's ability to identify magnets, classify objects as magnetic or non-magnetic, describe the main characteristics of magnets, and explain at least two uses of magnets correctly in daily life.*

Materials: Magnet, different wooden materials, plastic materials, glass utensils, stones, metallic objects, screw drive, papers, glass, wood, plastic, coin of 100, metallic door, metallic barrels, staples, ATM cards or national identity cards, a big radio set/ receiver, CD, flash disk, radio tapes, nails, paper clips, coins, iron keys, metal bottle caps, plastic spoon, rubber band, wood piece, paper, fabric, eraser, charts or posters showing types of magnets, uses of magnets in daily life, compass, etc..

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8. APPENDIX

Subjects and weekly time allocation for double shift in lower primary (P1-P3)

#	Subjects to be taught in P1-P3	Number of periods (Period = 40 minutes)
1	Kinyarwanda/ Rwanda Sign Language	5
2	English	5

3	Mathematics	5
4	French	2
5	Social Studies and Religious Education	2
6	Science and Elementary Technology (SET)	2
7	Creative Arts (fine arts and Music)	2
8	Physical Education and Sports	1
9	Co-Curricular activities	1
	Total number of contact periods per week	25
	Total number of contact hours/Week	16.6
	Total number of contact hours per year (39 weeks)	647.4